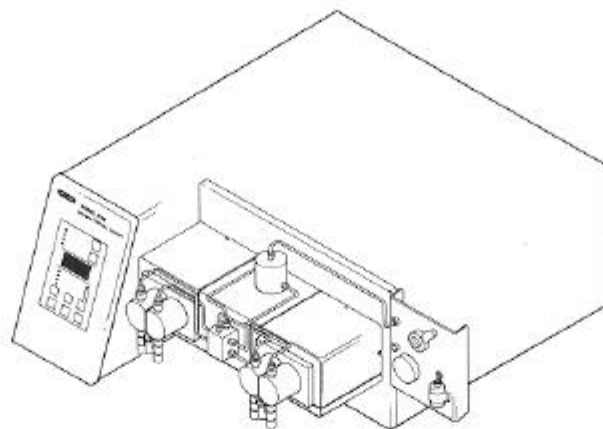




Model 2800 Solvent Delivery System

User's Manual

**Catalog Number
125-1527**

For Technical Service
Call Your Local Bio-Rad Office or
in the U.S. Call **1-800-4BIORAD**
(1-800-424-6723)

BIO-RAD

Table of Contents

	Page
Section 1 Introduction	1
1.1 Unpacking.....	2
Section 2 Set-up/Power-up Procedure.....	2
2.1 Location.....	2
2.2 Fuse Insertion.....	2
2.3 Electrical Connections.....	3
2.4 Liquid Connections	5
Section 3 Principles of Pump Operation	6
Section 4 Soft-Start® Flow Rate Ramping	7
Section 5 Pump Operation.....	7
5.1 Priming and Purging the System.....	7
5.2 Setting the Flow Rate	8
Section 6 Connecting an HPLC Column	9
Section 7 Setting Pressure Limits.....	9
Section 8 Compressibility Compensation	9
Section 9 Changing Solvents.....	10
Section 10 Washing the Pistons During Operation.....	10
Section 11 Maintenance.....	11
11.1 Lubrication	11
11.2 Changing the Seals	12
11.3 Changing the Pistons.....	14
Section 12 Troubleshooting Guide	14
Section 13 Troubleshooting Procedures	15
13.1 Troubleshooting Plan	15
13.2 Check Valves.....	15
13.3 Cleaning Check Valves.....	16
13.4 Seals and Pistons	16
13.5 Adjusting the Zero of the Pressure Display.....	16
Section 14 Replacement Parts	17
Section 15 Computer Interface	18
Section 16 Model 2800 Specifications.....	18

Warranty

Bio-Rad Laboratories warrants the Model 2800 Solvent Delivery System against defects in materials and workmanship for 1 year. If any defects occur in the instrument during this warranty period, Bio-Rad Laboratories will repair or replace the defective parts free. The following defects, however, are specifically excluded:

1. Defects caused by improper operation.
2. Repair or modification done by anyone other than Bio-Rad Laboratories or an authorized agent.
3. Use of fittings or other spare parts supplied by anyone other than Bio-Rad Laboratories.
4. Damage caused by accident or misuse.
5. Damage caused by disaster.
6. Corrosion due to use of improper solvent or sample.

For any inquiry or request for repair service, contact Bio-Rad Laboratories after confirming the model and serial number of your instrument.

Model _____

Catalog No. _____

Date of Delivery _____

Warranty Period _____

Serial No. _____

Invoice No. _____

Purchase Order No. _____

Section 1 Introduction

The Model 2800 Solvent Delivery System is for use in biotechnology labs with most commonly used HPLC solvents. The pumps provide constant flow using a dual piston positive displacement design to insure dependable, reproducible gradients. The Model 2800 Solvent Delivery System requires a computer and Bio-Rad's HPLC software to perform gradient HPLC. Features of the Model 2800 Solvent Delivery System include:

- Soft-Start flow rate ramping for longer column life
- Built-in high pressure mixing for reproducible chromatography
- Balanced dual piston positive displacement solvent delivery for precise HPLC gradients
- Built-in high resolution data collection for use with Bio-Rad's Windows®-based HPLC software
- Automatic solvent compressibility compensation at any pressure or flow
- Durable construction for reliable long-term operation
- Titanium version available for expanded solvent compatibility
- User-selectable over- and under-pressure alarms
- Four programmable relay outputs for interfacing with other laboratory instruments
- Smooth flow rates up to 6,000 psi at 9.9 ml/min. Both titanium and stainless steel versions have audible alarms to indicate over- and under-pressure faults.

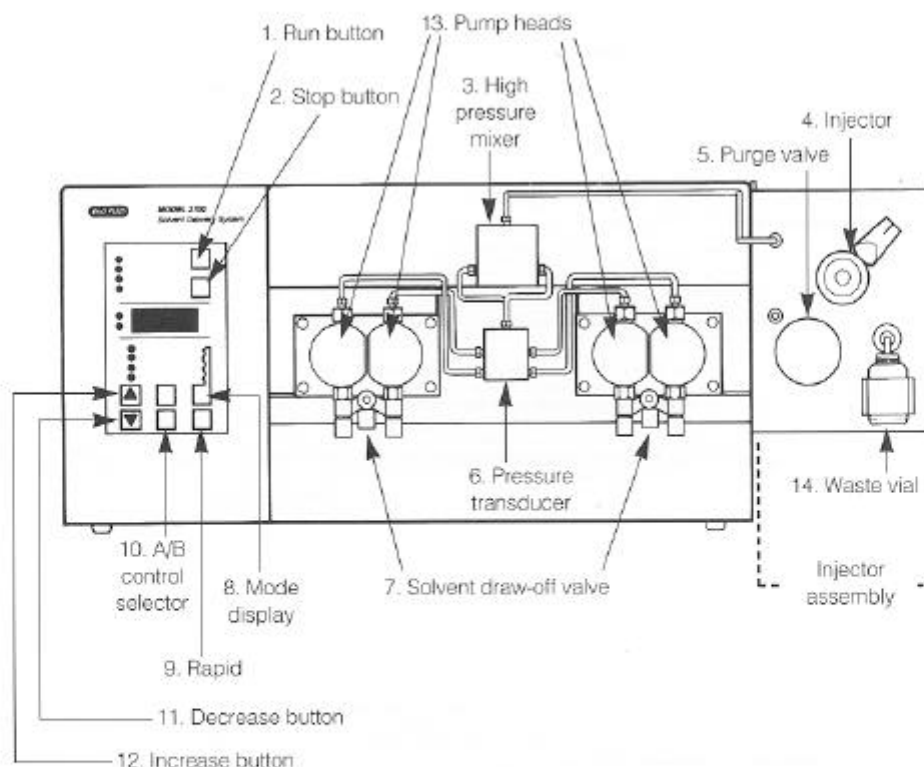


Fig. 1.1. Model 2800 Solvent Delivery System

1.1 Unpacking

Check the contents of your shipment against the following list of items. If anything is missing, call your local Bio-Rad office immediately and arrange shipment of replacement parts.

1. Warranty card
2. Spare fuse
3. Power cord
4. Hex Allen key, 2 sizes
5. Teflon tubing, 12 feet
6. Gland seals, 4
7. Seal plunger
8. Seal insertion guide
9. $\frac{1}{8}$ inch nut and ferrule, 2
10. Washing port tubing and fittings (titanium version only)
11. Spare seals, 2

Section 2 Set-up/Power-up Procedure

2.1 Location

The Model 2800 Solvent Delivery System operates at atmospheric pressure, and at temperatures between 5 °C and 40 °C. Allow 2 inches of clearance behind the instrument for proper ventilation.

2.2. Fuse Insertion

Units are normally shipped from Bio-Rad with the correct fuses for your area installed. Before operating the pump, look at the power entry module on the rear panel and note the voltage selection. If the voltage is correct for your area, proceed to the next section. To change the fuse:

1. Open the fuse cover using a small screwdriver.
2. Carefully remove the voltage selection card from the housing by pulling it straight out. Do not damage the plastic indicator pin or the metal contacts on the selector card.
3. The indicator pin can be moved across the slot on the voltage selector card. To set the required voltage for the pump, the indicator pin must be oriented on the side opposite the printed voltage on the voltage selection card (see Figure 2.1). In addition, the indicator pin must slide into the notch on the outside edge of the voltage selection card.
4. Insert the voltage selection card into the housing, with the edge reading the desired voltage first, and the printed side facing the AC plug and the on/off switch.
5. The white arrow on the indicator must align with the black arrow on the fuse housing block.
6. Replace the cover.

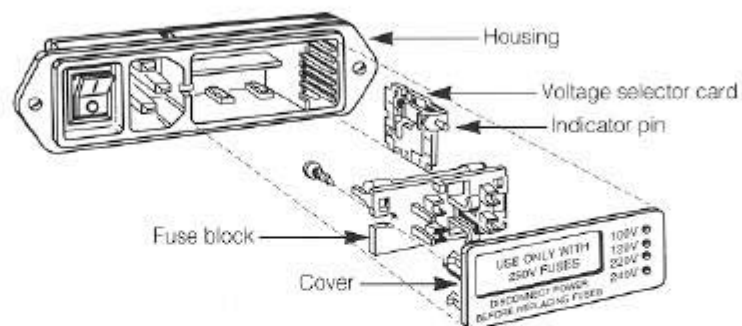


Fig. 2.1. Power entry module.

2.3. Electrical Connections

Connecting to a Computer

Connect the RS-232 cable from the Model 2800 connector labeled RS-232 port 1 to the RS-232 port on your computer. Use the null modem cable supplied with the Model 2800 Solvent Delivery System (part number 125-1642). To configure the RS-232 port on your computer, see the HPLC software manual.

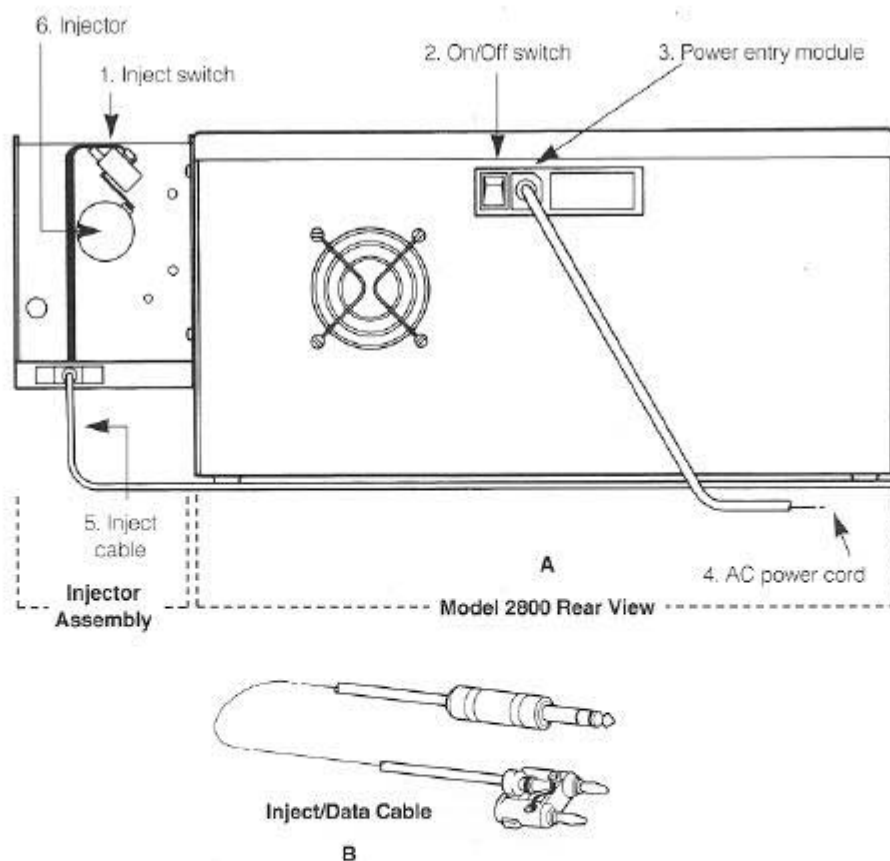


Fig. 2.2. Electrical connections.

Connecting Detectors

Connect the detectors to the inputs labeled Data channel 1 or data channel 2. The normal setting for these inputs is 1 volt full scale. For best results, use the integrator output or the 1 volt recorder output on the back of your HPLC detector.

Setting Video Strip Chart Input

To change the input voltage range of the data channel inputs, refer to Figure 2.3. Each data channel input can be set to either 10 mV, 100 mV, or 1 volt input. This input voltage corresponds to 100% on the video strip chart. The Model 2800 Solvent Delivery System comes from the factory set for a 1 volt input. This is compatible with integrator outputs on HPLC detectors. It is also used for recorder outputs if the HPLC detector can be set for a 1 volt setting. For most HPLC systems, the factory settings should be used.

To change the input voltage setting:

1. Remove the top cover of the Model 2800 Solvent Delivery System by removing the retaining screws on the side of the unit. Gently lift the top cover straight up and off the Model 2800 Solvent Delivery System.
2. Locate the electronics enclosure in the rear right side of the unit. Find the analog-to-digital circuit board. The analog-to-digital circuit board has switches along the card edge. Do not remove any connecting cables.
3. Set the switches according to Figure 2.3.



Board as viewed from edge.

Switch 1 Range Settings				
Input Range	Channel 1 or 3		Channel 2 or 4	
	SW1A	SW1B	SW1C	SW1D
1 V	☐ UP	☐ UP	☐ UP	☐ UP
100 mV	☐ Down	☐ UP	☐ Down	☐ UP
10 mV	☐ UP	☐ Down	☐ UP	☐ Down

Fig. 2.3. Analog input range selection.

Connecting the Injector

The injection valve can be mounted along either side of the Model 2800 Solvent Delivery System or along the top. For best results, mount the injection valve in a position that will minimize the tubing length between the output of the column and the input to the detector. Connect the switch on the injector (red and black terminals) to the inputs labeled Inject on the the Model 2800 Solvent Delivery System using the cable provided (part number 800-6126).

Power Connections

Before connecting any cords or cables, be sure the power switch is turned off. The power switch is in the top center on the back on the Model 2800 Solvent Delivery System. Insert the power cord into the rear panel jack of the Model 2800 Solvent Delivery System. Use an outlet with a grounding plug. To turn on the pump, switch the toggle switch on the power input module from 0 to 1.

Connecting a Fraction Collector

Bio-Rad's Model 2110 Fraction Collector connects directly to the Model 2800 Solvent Delivery System using the fraction collector control cable (catalog number 125-1658). Connect the 9 pin end of the cable to the Model 2110 Fraction Collector, and the 25 pin end to the Model 2800 port labeled Digital I/O. When connected, the fraction collector can be controlled using Bio-Rad's HPLC software. The software instruction manual will describe configuring the fraction collector.

Connecting a Model AS-100 Automatic Sampler

The AS-100 Automatic Sampling System connects to the Model 2800 Solvent Delivery System via the connector labeled Automatic Sampler using Bio-Rad cable (catalog number 125-1654). With the AS-100 Automatic Sampling System connected, the Model 2800 Solvent Delivery System can signal the AS-100 Automatic Sampling System to start in injection sequence. See the AS-100 Automatic Sampling System manual and Bio-Rad HPLC software manual for details on operating the AS-100 Automatic Sampling System.

2.4 Liquid Connections

Refer to Figure 2.4 for details about plumbing the Model 2700 Solvent Delivery System.

1. Connect one end of the HPLC tubing, using the nut and ferrule supplied, to the outlet of the high pressure mixer assembly as shown in Figure 2.4. Use either Fingertight fittings or regular nuts and ferrules. The stainless fittings withstand 6,000 psi, and the PEEK fittings are capable of 4,000 psi.
2. The other end of the tubing connects to the injector assembly inlet. The injector assembly may be mounted next to or on top of the Model 2800 Solvent Delivery System. See Figure 11.1 for mounting details.

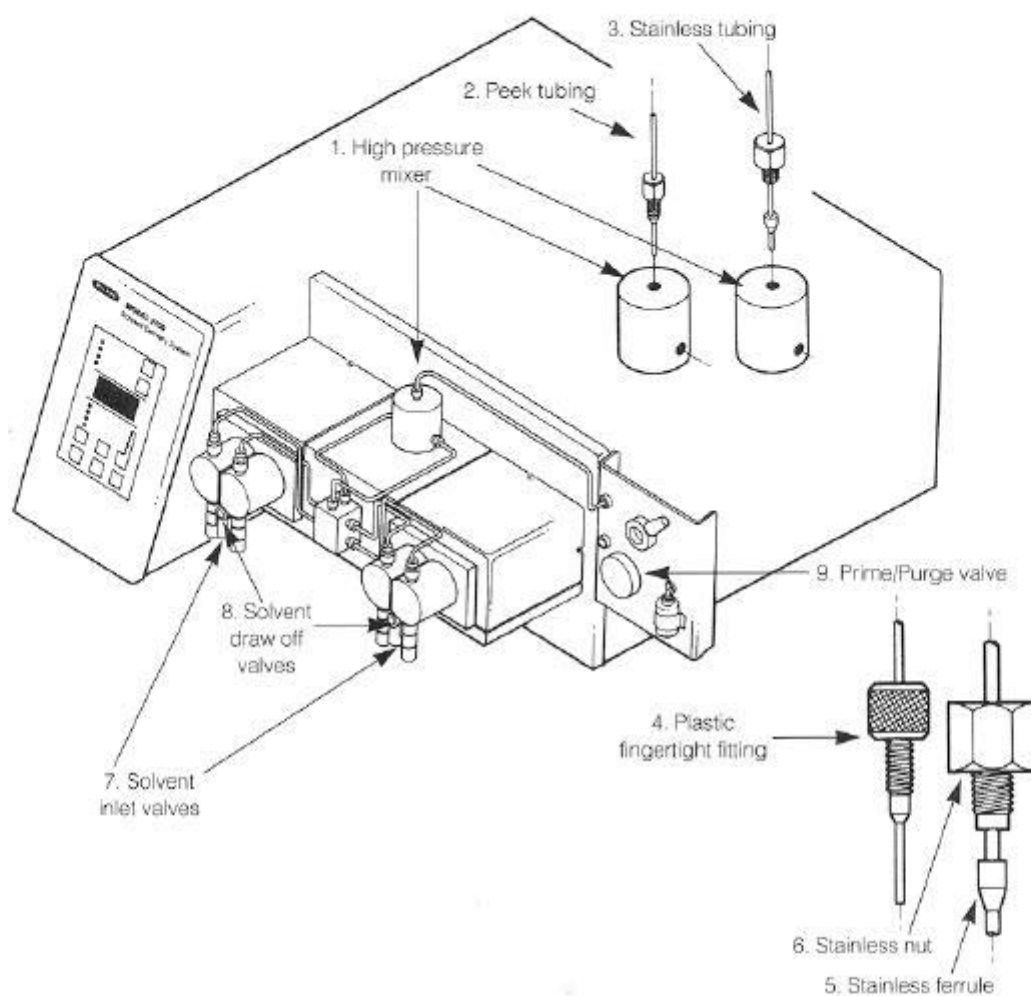


Fig. 2.4. Model 2800 Solvent Delivery System liquid connections.

Section 3

Principles of Pump Operation

The Model 2800 Solvent Delivery System uses two dual piston pumps and a high pressure mixer to provide accurate, reliable operation. The pistons operate in tandem to provide solvent delivery with a minimum of pump pressure pulsation.

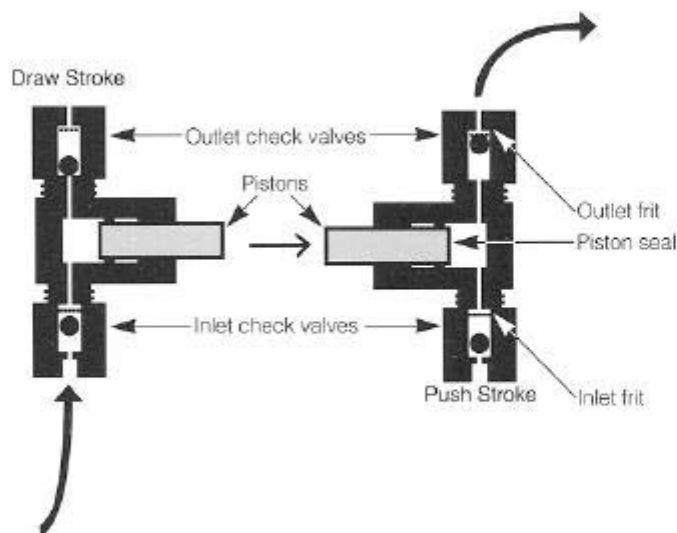


Fig. 3.1. Pump operation.

Figure 3.1 is a side view schematic of one of the pump heads in the Model 2800 Solvent Delivery System. During the draw stroke, the inlet check valve opens and provides a path for solvent to enter the cylinder. The outlet check valve closes to prevent solvent from being sucked back into the cylinder after it has been displaced.

During the push stroke, the outlet check valve opens, and the inlet check valve closes. The piston pushes solvent from the cylinder into the high pressure dynamic mixer. Solvent flow rate is determined by the speed of the pistons.

Section 4

Soft-Start Flow Rate Ramping

Most HPLC pumps are designed to reach the programmed flow rate quickly. For HPLC columns, rapid ramping flow rates cause rapid pressure increases creating a sudden shock to the column. For most applications, a gradual increase in pressure will extend the column life. The Model 2800 Solvent Delivery System has a special Soft-Start feature to prevent column shock. When the pump is started, it gradually increases the flow rate to the final programmed value. After the 1.5 minute Soft-Start period, any flow rate changes occur immediately, so the system can produce any gradient profile using an external controller.

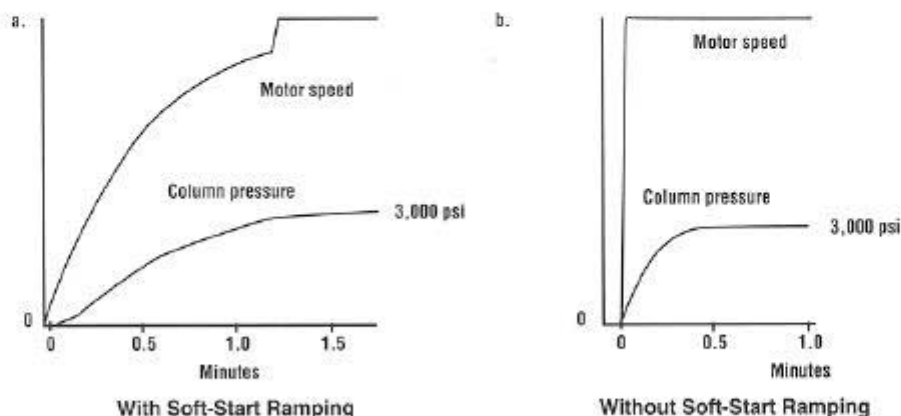


Fig. 4.1. Soft-Start pumping.

Section 5

Pump Operation

5.1 Priming and Purging the System

Priming your HPLC system is a two step process. First you must remove any air bubbles from the solvent inlet lines going into the pump. Then you must remove any air inside the HPLC tubing leading to the column.

Purging the solvent inlet lines

1. Place the tubing from the solvent inlet valves (Figure 2.3, item 7) into a reservoir of an appropriate solvent. Isopropanol, methanol, and ethanol are most effective. Water also works well. The pump was factory tested using isopropanol, and the unit is shipped containing isopropanol in the flow path. Therefore isopropanol or one of the recommended isopropanol-miscible solvents should be used for initial priming. Maintain the solvent reservoir at an elevation higher than the pump heads to maintain hydrostatic pressure on the inlet check valves.
2. Use the draw-off valves (Figure 2.3, item 8) to pull solvent through the inlet tubing. Place a syringe into the fitting, turn the valve fully counter clockwise, and draw solvent into the syringe. Draw solvent through the system until no air bubbles enter the syringe.

Priming the system

1. Connect the tubing directly into the HPLC detector flow cell without a column in-line.
2. Turn the prime/purge valve (Figure 2.3, item 9) on the injector assembly completely counterclockwise.
3. Allow solvent to run at 5 ml/min for 5 minutes. (See Section 5.2, Setting the Flow Rate.) This will clear air bubbles from the pump heads and pressure transducer.
4. Place the outlet end of the discharge tubing into a suitable waste container. Check the discharge tubing for consistent solvent flow. If the flow fluctuates, repeat step 3.
5. Turn the prime/purge valve fully clockwise and allow solvent to flow into the detector flow cell. This will clear air bubbles in the flow cell. After all air bubbles are cleared, connect the column.

5.2 Setting the Flow Rate

If the HPLC Software is running the Model 2800 can only be operated remotely. For local operation turn off the computer or exit the HPLC software.

1. Turn on the pump using the switch on the rear panel (Figure 2.2, item 2). The display should indicate Stop Flow and give a flow rate of 0 ml/min.
2. Push RAPID and INCREASE (see Figure 5.1) until the display reads 1.0 ml/min, or 4.0 ml/min for the 40 ml/min systems. If you exceed this flow rate, use INCREASE and DECREASE to re-adjust the flow rate to 1.0 ml/min.
3. Push RUN. The stop flow lamp should extinguish, and, after the pump starts, the pumping lamp should light. To stop the pump, push STOP.

Note: The Soft-Start feature will gradually increase the flow rate.

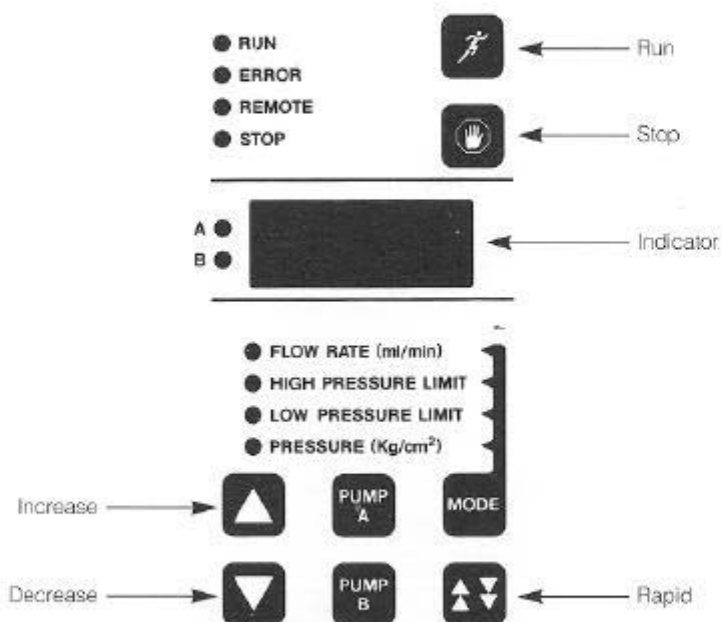


Fig. 5.1. Model 2800 Solvent Delivery System control panel.

Section 6 Connecting an HPLC Column

After the system is purged of air, connect the HPLC column. Be sure to check the correct fluid flow direction for the column you are using. Some columns can be run in either direction, and some must only operate in the fluid direction recommended by the manufacturer.

The column connects to the outlet fitting on the injector labeled column. Use a short length of narrow bore tubing to connect the injector to the column. Short lengths of tubing will minimize sample dispersion and peak broadening. Connect the column to the detector flow cell with a short length of tubing. The output from the flow cell should be fed to a suitable drain for the solvents you are using.

Section 7 Setting Pressure Limits

The pressure limit circuits stop the pump if the pressure exceeds the user-selected high or low pressure. The high and low pressure limits are set on the front panel. These pressure limits keep excess pressure from damaging valves or columns, or system leaks from damaging the pump or creating a safety hazard.

Note: Both the high and low pressure limits can be set anywhere in the 0-445 kg/cm² range (or 0-125 kg/cm² for the titanium systems). The low pressure limit is not activated until the pressure exceeds the setpoint. This prevents low pressure during flow ramping from triggering a low pressure alarm. If the low pressure limit is set above the high pressure limit, it is never activated.

1. Set the high and low pressure limits by using SELECT to choose either the High Pressure Limit or the Low Pressure Limit display. (See Figure 5.1.) Use INCREASE or DECREASE, along with RAPID, to enter the limits.
2. If the high pressure limit is reached, the alarm sounds, the unit stops, and the high pressure indicator flashes. Press STOP, wait for the pressure to drop, eliminate the cause of the over-pressure, then press RUN to restart.
3. When the pressure falls below the low pressure limit for 30 seconds, the alarm will sound, the pump will stop, and the low pressure limit indicator will flash. Press STOP. After correcting the problem, press RUN to restart the pump.

Section 8

Compressibility Compensation

The Model 2800 Solvent Delivery System is factory calibrated to provide accurate solvent delivery. The compressibility compensation circuit maintains accuracy despite changes in operating pressure. This compensation cannot be adjusted manually.

Section 9

Changing Solvents

Usually switching from one solvent to another is a simple operation. However, some solvents do not mix well with others, creating bubbles, two liquid phases, or precipitates. These conditions can be avoided by carefully selecting a sequence of solvents that will be compatible with the HPLC system. Sometimes a gradual increase in the concentration of a new solvent can prevent problems. This is the case with water and methanol. If you do not know what happens when you mix the solvents, mix the solvents in a beaker and observe the reaction before mixing them in the HPLC pump.

Section 10

Washing the Pistons During Operation

The Model 2800 Solvent Delivery System includes a feature which allows the user to periodically rinse the back of the pistons and seals without removing any parts. On the top and bottom of each of the heads there is a 10-32 threaded hole for introducing water or other solvents which rinse the back of the piston and the seal. This procedure is normally not required; however, rinsing removes small amounts of salts or other aggressive solvents that may have passed through the seals.

Accumulated salts on the piston cause piston scoring as well as accelerated seal wear. Regular rinsing helps to prolong the life of the seals and the pistons. See Figure 10.1. Under most conditions, flushing with 20 ml of water every 2 hours will prevent precipitates from accumulating on the piston. In extreme conditions, wash every one-half hour or continuously via a gravity-fed siphon.

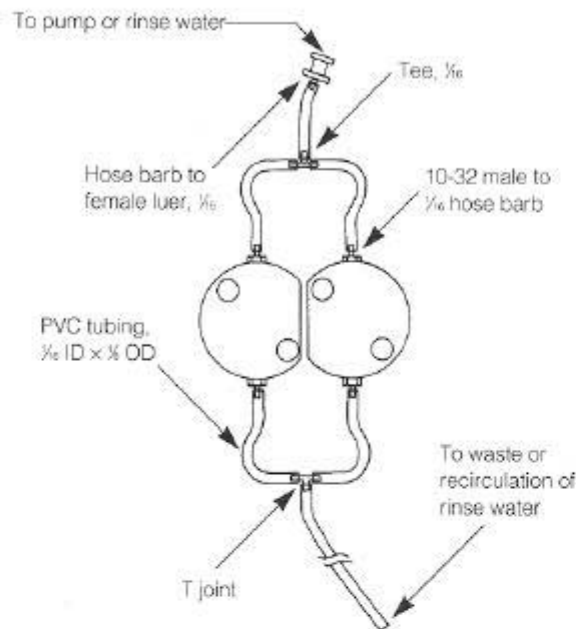


Figure 10.1. Plumbing the piston washing accessory.

Section 11 Maintenance

11.1 Lubrication

Lubricate every 3 months or after every 500 hours of operation.

1. Remove the pump enclosure top (Figure 11.1, item 1) by removing the case screws and lifting off the cover.
2. Use a paper towel to wipe the cam surface and cam followers (Figure 11.1, item 15). This will remove some of the old lubricant.
3. Apply 10 drops of 90 W motor oil or other lightweight all-purpose oil to the cam wiping sponge.
4. Apply 2 drops of 90 W oil to the cam follower bearings.
5. Apply 2 drops of 90 W oil to the cam followers.
6. Replace the pump enclosure and re-install the screws.

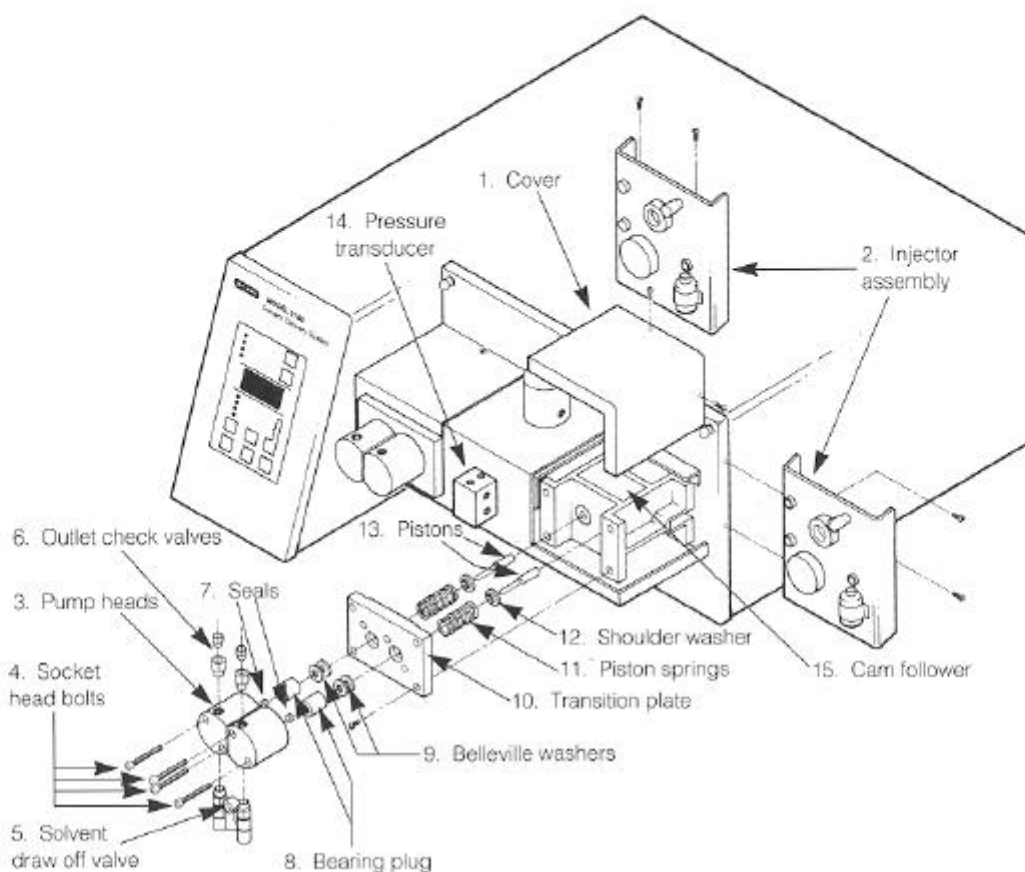


Fig. 11.1. Exploded view of the Model 2800 Solvent Delivery System.

11.2 Changing the Seals

Before changing the pump seals, refer to Figure 11.2 and familiarize yourself with the mechanical details of the pump head.

1. For best visualization of the cam, cam followers, and pistons, remove the enclosure cover by removing the case screws and lifting off the cover (Figure 11.1, item 1).
2. Disconnect the solvent draw-off valve (Figure 11.1, item 5) and outlet tubes from the outlet check valves. Remove the inlet and outlet check valves.

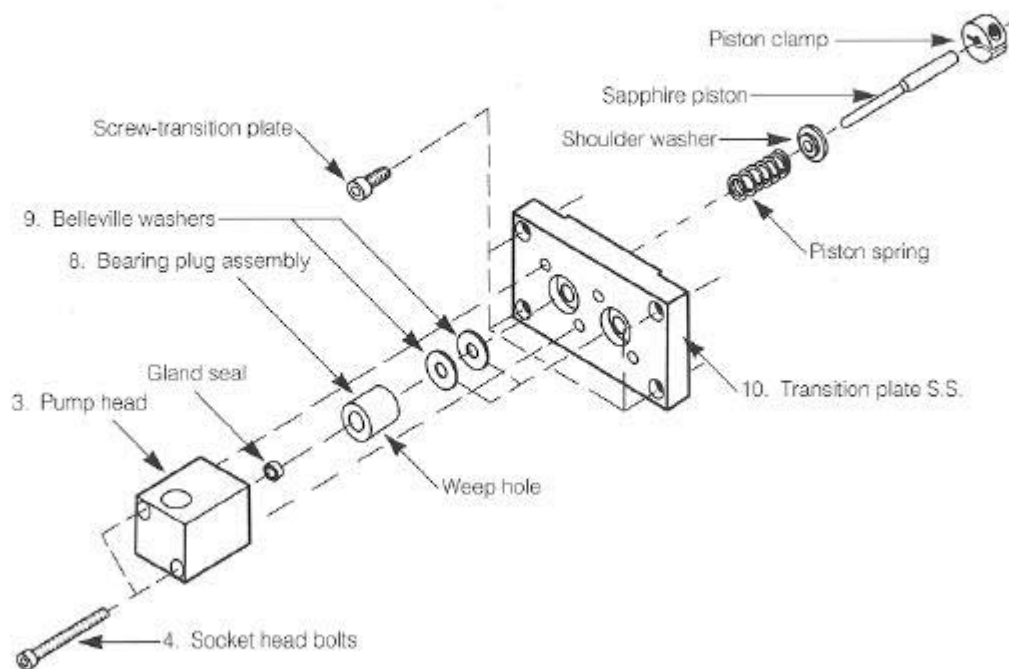


Fig. 11.2. Exploded view of piston assembly.

3. Remove the two socket head bolts (Figure 11.2, item 4) which secure the pump head to the transition plate (item 10). Use the $\frac{5}{16}$ " hex wrench provided with the pump.
4. Gently remove the pump head (Figure 11.2, item 3) by pulling forward slowly until the piston is free of the high pressure seal. Pull directly out from the transition plate, since angular loads will break the piston.

Caution: Do not touch the pistons with your bare hands. Fingerprints can damage the seals and score the pistons.

5. Remove the bearing plug as shown in Figure 11.2, item 8, and set it aside. The belleville washers (item 9) are between the bearing plug and the face plate. Remove them and note the direction of the belleville washers for proper orientation on re-assembly.

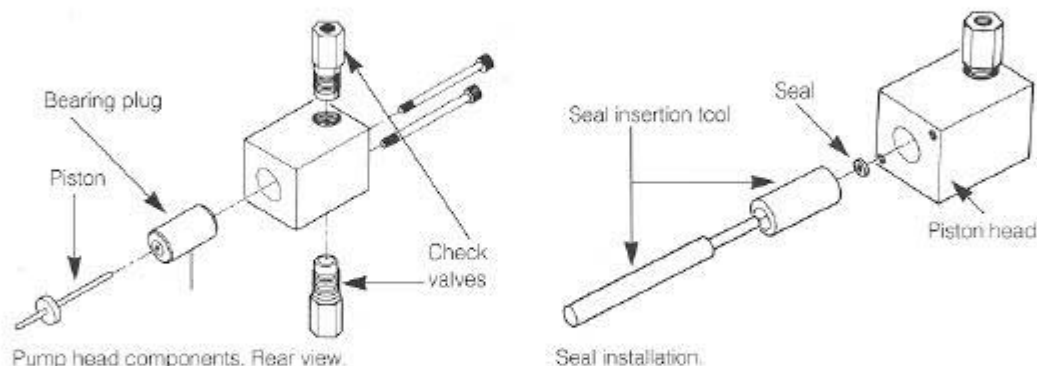


Fig. 11.3. Piston seal insertion.

6. Remove the old seal from the pump head. Be very careful not to scratch any of the metal surfaces when removing the seals. Discard the old seal. Clean the seal cavity, bearing plug, and belleville washers with alcohol or soapy water. Do not use abrasive cleaners. Be careful not to scratch the bearing surface in the bearing plug.
 7. Use the seal insertion tool (provided with the pump) to install the new seal as shown in Figure 11.3. One side of the seal has a spring visible. Make sure the spring faces the front (high pressure) side of the piston head. Be sure that the seal bottoms squarely into the seal cavity.
 8. Reinstall the two belleville washers over the piston shaft, and into the transition plate, in the correct orientation.
 9. Run the pump slowly until the exposed piston is fully extended. Clean the piston with alcohol and dry the piston. Slide the bearing plug back onto the piston shaft. Gently work the plug onto the shaft until the plug is fully seated against the transition plate. The flush end of the bearing plug should be facing out.
- Caution:** The weep hole in the bearing plug surface should be pointed directly down (10 ml/min version only). Do not touch the piston with your bare hands.
10. Wet the new seal with isopropyl alcohol. Re-install the pump head onto the pump, pushing very gently and straight toward the transition plate until the sapphire piston becomes inserted into the new seal. Push the pump head straight back to the transition plate.
 11. Replace the two socket head bolts in the pump head and tighten alternately $\frac{1}{4}$ turn until there is no gap between the pump head and transition plate.
 12. Reinstall the inlet check valve and plumbing. Prime the pump with methanol or isopropanol. (See Section 5.1.) Reinstall the outlet check valves and run for about 2-6 hours with 150-200 kg/cm² (10 ml/min) or 25-50 kg/cm² (40 ml/min) pressure to break in the new seal.

11.3 Changing the Pistons (Refer to Figures 11.1 and 11.2)

1. Remove the pump head as described in Section 11.2.
 2. Rotate the cam until the piston reaches its rear-most position.
 3. Place the piston centering tool over the piston.
 4. Insert the piston fork between the collar and the cam follower.
 5. Loosen the collar cap screw with the Allen wrench.
 6. Remove the piston with the piston centering tool. Do not drop the piston.
- Note:** Do not remove the piston fork until a new piston is in place.
7. Insert a new piston by following steps 1-6 in reverse order.

Section 12

Troubleshooting Guide

Table 1 contains a guide showing solutions to several pump related problems. The solutions listed below are described in Section 13, Troubleshooting Procedures.

Table 1. Troubleshooting Guide

Symptom	Cause	Solution
1. Pump does not run	a. Power off b. Pump stopped c. Flow set to zero d. Pressure limits set incorrectly e. Motor problem	a. Check power cord, fuse b. Push RUN c. Increase flow rate d. Check pressure limits and reset e. Contact Bio-Rad's Instrument Service
2. Pressure fluctuations	a. Some is normal b. Failed seal c. Failed inlet check valve d. Failed outlet check valve e. Stuck piston f. Broken or scratched piston	a. Check pressure against troubleshooting plan—Section 13.1 b. Replace seals c. Replace valve d. Replace valve e. Remove and clean f. Replace pistons
3. Leaks	a. Loose fittings b. Worn fittings c. Worn seals (leak at back of pump head)	a. Tighten fittings b. Replace fittings c. Replace seals
4. Seals wear out often	a. Clogged outlet frits and seals	a. Replace check valve, filter solvents (0.45 μ m filter)
5. Corrosion	a. Exposure to halides and low pH	a. Avoid halides when low pH is being used
6. Loud noises when the pump is running	a. Poor lubrication	a. Lubricate cam followers

Section 13

Troubleshooting Procedures

Introduction

There are three main reasons for pump failures, air bubbles, faulty check valves, and worn piston seals. Often problems in chemistry or detectors initially appear as a pump problem. When you first purchase your HPLC system you should establish a test procedure that rapidly determines which system component is at fault.

13.1 Troubleshooting Plan

The easiest troubleshooting is a process of elimination. Each component can be tested individually before failures occur. When problems develop, eliminate each component as a

source of failure until the faulty component is found. When the system is new, the following steps should be taken to develop a troubleshooting procedure:

1. Remove the column from the system and record the normal backpressure. Replace the column with a zero dead volume connector to take the pressure reading. The normal system pressure should include all system components, including detectors.
2. Without a column connected, perform a gradient linearity profile. Run a linear gradient from 100% water to 100% acetone over 20 minutes. Any normal UV absorber will suffice if acetone is not available. Use 1.5 ml/min as the flow rate and 220 nm for the detector, if it is available. Solvents must be well degassed before running this test.

From this chromatogram, you can measure the total system dead volume (typically less than 2 ml) and the linearity of the gradient. Keep this chromatogram as part of the troubleshooting procedure. If retention time problems develop in the future, run this test to eliminate the HPLC pumps as the source of the problem.

3. Remove the detector from the fluid flow path, dry out the flow cell, and perform a drift and noise test. This must be done with a dry flow cell to prevent any pump noise from affecting detector performance. To dry the flow cell, flush with isopropyl alcohol and dry with dry nitrogen. Drift and noise specifications are included in the detector manual. Record the results as part of the troubleshooting plan.
4. Most Bio-Rad HPLC columns are supplied with a test standard. The standard should be run for each new column and the results stored as part of the troubleshooting procedure. Whenever you suspect problems with the HPLC column, use the test mixture to eliminate the HPLC column as the source of the problems.

Before troubleshooting the pump, become familiar with the pump operation section of this manual—Section 3.

13.2 Check Valves

Excessive pressure pulsations and flow inaccuracies are usually from poorly functioning check valves. Most check valves fail to seat because of trapped air. Unfiltered solvents can also cause the valve to malfunction. Overtightening the check valves against the pump heads may crack the internal parts of the check valve. Finally, the linear velocities of the fluid within the check valve are so great that actual erosion of the balls and seats can occur after extended operation.

Another common cause of check valve failure is clogging of the frit on the outlet check valves. Clogging can be caused by unfiltered solvents, or bacterial or fungal contamination. If the frits are clogged, the pump continues operating with high pressure and excessive seal wear results. Check the system against the pressure recorded in the troubleshooting procedure.

In extreme cases, the clogged outlet check valves cause the seals to fail frequently and the pump heads feel warm when the pump is operating.

To purge the pumps of air, loosen the outlet tubes of each outlet check valve. Remove the two outlet check valves. With the solvent bottle elevated slightly above the pump, adjust the flow rate to 7 ml/min and run the pump for about 30 seconds to allow fluid to pass through the inlet check valves and pump heads. Re-install the outlet check valves and the outlet tubes.

13.3. Cleaning Check Valves

The check valves act to prevent fluid from flowing in the wrong direction. See Section 3, Pump Operation. If the inlet check valve becomes dirty, the ruby ball may not seat properly

and the pump may leak fluid back toward the solvent reservoir. To prevent this, the inlet check valves may need periodic cleaning.

If an outlet check valve becomes dirty, the ruby ball can be prevented from seating properly and leaks can develop. In addition, the outlet check valve is equipped with a frit (or filter) that prevents particles from being introduced into the sample or column. If the frit becomes clogged, pressure can build up in the piston chamber causing unnecessary seal wear. To prevent this, eluants should always be filtered using 0.45 μm filters (or smaller) immediately before use. Also, the outlet check valve should be cleaned periodically.

To clean the check valves, use an ultrasonic cleaner. Sonicate first in deionized and filtered water, then in 1% sodium dodecyl sulfate (SDS)-water, then water, and finally isopropanol. Clean in each solvent for 15 minutes. The outlet check valves must be replaced if insoluble particles are lodged in the frits.

13.4 Seals and Pistons

Seals usually wear out before pistons, but both give similar symptoms. In both cases, there is some leakage of fluid on the underside of the pump head. There is a small hole there to allow leaking fluid to be released if a problem occurs. If these conditions exist, replace the seals as shown in Section 11.2. If changing the seals does not correct the problem, replace the piston as described in Section 11.3.

13.5 Adjusting the Zero of the Pressure Display

Slight drift of the zero pressure readout will occur with temperature variations or mechanical stress caused by loosening or tightening the tubing fitting on the pressure transducer. The zero display can be re-adjusted by turning the pressure adjustment screw under the pump heads. The two adjustment screws vary the zero reading on the display.

Section 14

Replacement Parts

Here is a list of replacement parts and their corresponding catalog numbers. If you are unsure of the part you need, call our instrument service department.

100-2120	Bearing Plug
100-2130	Bearing Plug, Titanium
100-2390	Board Assembly, Display
100-2750	Board Assembly, Main Circuit
100-3420	Cam Follower Assembly
100-3570	Check Valve Rebuild Kit, Inlet, Stainless Steel
100-3580	Check Valve Rebuild Kit, Inlet, Titanium
100-3590	Check Valve Rebuild Kit, Outlet, Stainless Steel
100-3600	Check Valve Rebuild Kit, Outlet, Titanium
100-5910	Motor
100-5990	Nut, Solvent Draw-off Valve
100-6030	O-Ring, Pressure Sensor
100-6230	Piston Clamp
100-6240	Piston Washing Kit, Titanium
100-7010	Pump Head, 10 ml, Stainless Steel
100-7020	Pump Head, 10 ml, Titanium
100-7380	Screw, Pump Head, 4
100-7390	Screw, Transition Plate, 4
100-7470	Sil Pad
100-7590	Spring, Piston, 2
100-8880	Washer, Belleville, 4
100-8890	Washer, Shoulder
125-0302	Piston Seals, stainless, 2
125-0303	Piston Seal Insertion Tool, 10 ml
125-0304	Piston, 10 ml
125-0305	Piston Insertion Tool, 10 ml
125-0306	Check Valve, Inlet, Stainless Steel
125-0307	Check Valve, Outlet, Stainless Steel
125-1348	Solvent Draw Valve
125-1353	Piston Seals, 2, Titanium
125-1356	Check Valve, Inlet, Titanium
125-1357	Check Valve, Outlet, Titanium

Section 15

Computer Interface

The computer interface is a 9 pin RS-232 connector located on the right side of the Model 2800. Bio-Rad's HPLC software must be used to control the instrument remotely.

Section 16

Model 2800 Specifications

Power requirements	100, 120, 220, 240 volts 50/60 Hz, 200 W
Type	Positive displacement dual piston pumping
Flow rates	0.02 to 9.9 ml/min in 0.01 ml/min increments (10 ml/min version)
Flow rate accuracy	1% or better
Flow rate reproducibility	0.1% or better
Operating pressure	0-420 kg/cm ² (10 ml) or 0-100 kg/cm ²
Temperature range	4-40 °C
Pressure limits	Adjustable in 1 kg/cm ² increments. Audible alarm indicates over or under pressure conditions.
Remote control input	0-10 volt DC input
Mixing	Built-in high pressure dynamic mixing
Detector inputs	2
Data input range	10 mV, 100 mV, & 1 V - user selectable
Sample Rate	10 Hz
Relay outputs	4 programmable

Windows is a trademark of Microsoft Corp.

BIO-RAD

Life Science Group

2000 Alfred Nobel Drive
Hercules, California 94547
Telephone (510) 741-1000
Fax: (510) 741-1060

Eastern Regional Office, P.O. Box 1229, 85A Marcus Dr., Melville, New York 11747 • Phone (516) 756-2575 • Fax (516) 756-2594
European Headquarters, Bio-Rad Laboratories, Dève du Senechal, 19, B-1180 Brussels • Phone 02 375 59 70 • Fax 02 374 61 62
Australia, Bio-Rad Laboratories Pty Limited, Unit 11, 112-118 Talavera Rd P.O. Box 371, North Ryde, N.S.W. 2113 • Phone 02-805-5000 • Fax 02-805-1920
Austria, Bio-Rad Laboratories Ges.m.b.H., Aufhofstrasse 78D, A-1130 Wien • Phone 0222-87 76 901 • Fax 0222-87 65 629
Belgium, Bio-Rad Laboratories S.A./N.V., Begoniastraat 5, B-9810 Nazareth Eke • Phone 091-85 55 11 • Fax 091-85 65 54
Canada, Bio-Rad Laboratories (Canada) Ltd., 5149 Brackley Boulevard, Mississauga, Ontario L4W 2A6 • Phone (416) 624-0713 • Fax (416) 624-3019
China, Bio-Rad Laboratories, Yanshan Hotel Office Tower, #1317, A138 Haidian Road, Beijing • Phone 2564308 • Fax 2566640
France, Bio-Rad S.A., 94/96 rue Victor Hugo, B.P. 220, 94203 Ivry Sur Seine Cedex • Phone 01-49 60 683 4 • Fax 01-46 71 24 67
Germany, Bio-Rad Laboratories GmbH, Heidemannstraße 164, Postfach 45 01 33, D-8000 München 45 • Phone 089-318 84-0 • Fax 089-318 84 100
Italy, Bio-Rad Laboratories S.r.l., Via Cellini, 18A, I-20090 Segrate Milano • Phone 02-21809.1 • Fax 02-21809-399
Japan, Nippon Bio-Rad Laboratories, K. K., Sumitomo Seimei Kachidoki Bldg 5-3-6 Kachidoki, Chuo-Ku, Tokyo 104 • Phone 03-3534-7515 • Fax 03-3534-8027
The Netherlands, Bio-Rad Laboratories B. V., Fokkerstraat 10, 3905 KV Veenendaal • Phone 08385-4066 • Fax 08385-42216
New Zealand, Bio-Rad Laboratories, Pty Ltd., P. O. Box 100-001, North Shore Mail Centre, Auckland 10 • Phone 09-443 3099 • Fax 09-443 3097
Pacific, Bio-Rad Laboratories, Unit 1111, 11/F., New Kowloon Plaza, 39, Tai Kok Tsui Road, Tai Kok Tsui, Kowloon, Hong Kong • Phone 7893300 • Fax 7891257
Scandinavia, Bio-Rad Laboratories, Kanalvägen 10C, 19461 Upplands Väsby, Sweden • Phone 46 (0) 760-73489 • Fax 46 (0) 760-71781
Spain, Bio-Rad Laboratories, S. A. Avda Valdeaparra 3, Pol. Ind. Alcobendas, E-28100 Alcobendas, Madrid • Phone (91) 661 70 85 • Fax (91) 661 96 98
Switzerland, Bio-Rad Laboratories AG, Kamalsstrasse, 17, CH-8152 Glattpfurg • Phone 01-810 16 77 • Fax 01-810 19 33
United Kingdom, Bio-Rad Laboratories Ltd., Bio-Rad House, Maylands Avenue, Hemel Hempstead, Herts HP2 7TD • Phone 0900 181134 • Fax 0442 259118